

AI-Powered Industrial Chatbot (Generative AI)

Title

AI-Powered Industrial Chatbot Using Hugging Face Large Language Models

1. Problem Statement

Design and implement an **AI-powered industrial chatbot** that:

- Uses a Hugging Face language model
- Answers questions from provided industrial/technical documents
- Applies prompt engineering or fine-tuning
- Provides a usable chatbot interface (CLI or web)

2. Tasks / Questions to Be Solved

Question 1: System Architecture

Question: Design the architecture of the chatbot showing document input, LLM, embedding/retrieval, and user interface.

Question 2: Model Selection

Question: Select a suitable Hugging Face LLM and justify its use for industrial Q&A tasks.

Question 3: Knowledge Ingestion

Question: Convert provided documents into a format usable by the chatbot (text chunks, embeddings, or prompts).

Question 4: Prompt Engineering / Fine-Tuning

Question: Implement prompt engineering or fine-tuning to improve domain-specific answers.

Question 5: Q&A System Implementation

Question: Build a question–answering pipeline that retrieves relevant document content and generates responses.

Question 6: Chatbot Deployment

Question: Deploy the chatbot as a command-line interface (CLI) or simple web application and demonstrate interaction.

3. Tools to Be Used

- **Python**
- **Hugging Face Transformers**
- **LangChain / LlamaIndex (optional)**
- **FAISS / ChromaDB (optional)**
- **Streamlit / Flask (for web UI)**

4. Submission Guidelines (Google Drive)

Upload the following to **Google Drive** and submit the shared link:

- **Project Report (PDF)**
- **Source Code Folder**
- **Sample documents used for Q&A**
- **Screenshots / screen recording of chatbot interaction**

Folder Name:

AI_Industrial_Chatbot_<StudentName>